Analytical Chemistry Cumulative Examination
December 6, 2013

In the November 19, 2013 issue of Analytical Chemistry a leading article was entitled Editors' Highlights. Its topic is "Single Cell Tracking of Gadolinium Labeled CD4+ T Cells . . ." Please answer the following questions about this paper.

1.. What sampling technique is used?

2. What is the primary analytical method employed?

3. To what other analytical method is it compared?

4. What are the main advantages (over other methods) described for the technique?

5. What detection limits are claimed and what is the prediction about possible improvements in those detection limits?

6. What type of labeling is described in this paper?

7. What is the time frame described for detection of the labeled cells?

8. How was the mouse study carried out? Explain the sequence of steps for this study. Be specific.